Unified native factor-base sweeper

Engineering implementation · 2026-10-17–18

One launch command enumerates structured bases, constructs exact native point inventories and polynomial systems, runs degree-bounded F4, verifies actual point decompositions, and preserves resumable numbered attempts. Human stderr lines follow the repository’s CADO-style stage conventions; durable JSONL has the same events.

Real-time feedback

The display separates verified decompositions, refutations, exhausted probes and independent training rank. It reports cumulative and rolling verified rates, independent relations per second, provisional phase/rank ETAs, and provisional ratios to matched prefixes of the previous and first completed baseline candidates on the same curve. Positive controls and holdout probes never enter training rank. Unknown estimates remain unknown.

Rates include finite-inventory truth census, system construction, solving, witness checks and incremental logging. Initial base/curve construction is separate. Shared-host telemetry is not isolated timing evidence. No total-cost winner or end-to-end speedup is admitted; native pricing remains incomplete.

Implemented scope and limits

Default functional fixtures cover small binary and prime fields. Seed and dimension grids are available without editing JSON. Binary recipes retain every basis word; exact ordered signed inventories retain FB1, FBM1, an order Merkle commitment and artifact hashes. Verification independently reconstructs inventories and replays point witnesses and relation rows; F4 is not rerun.

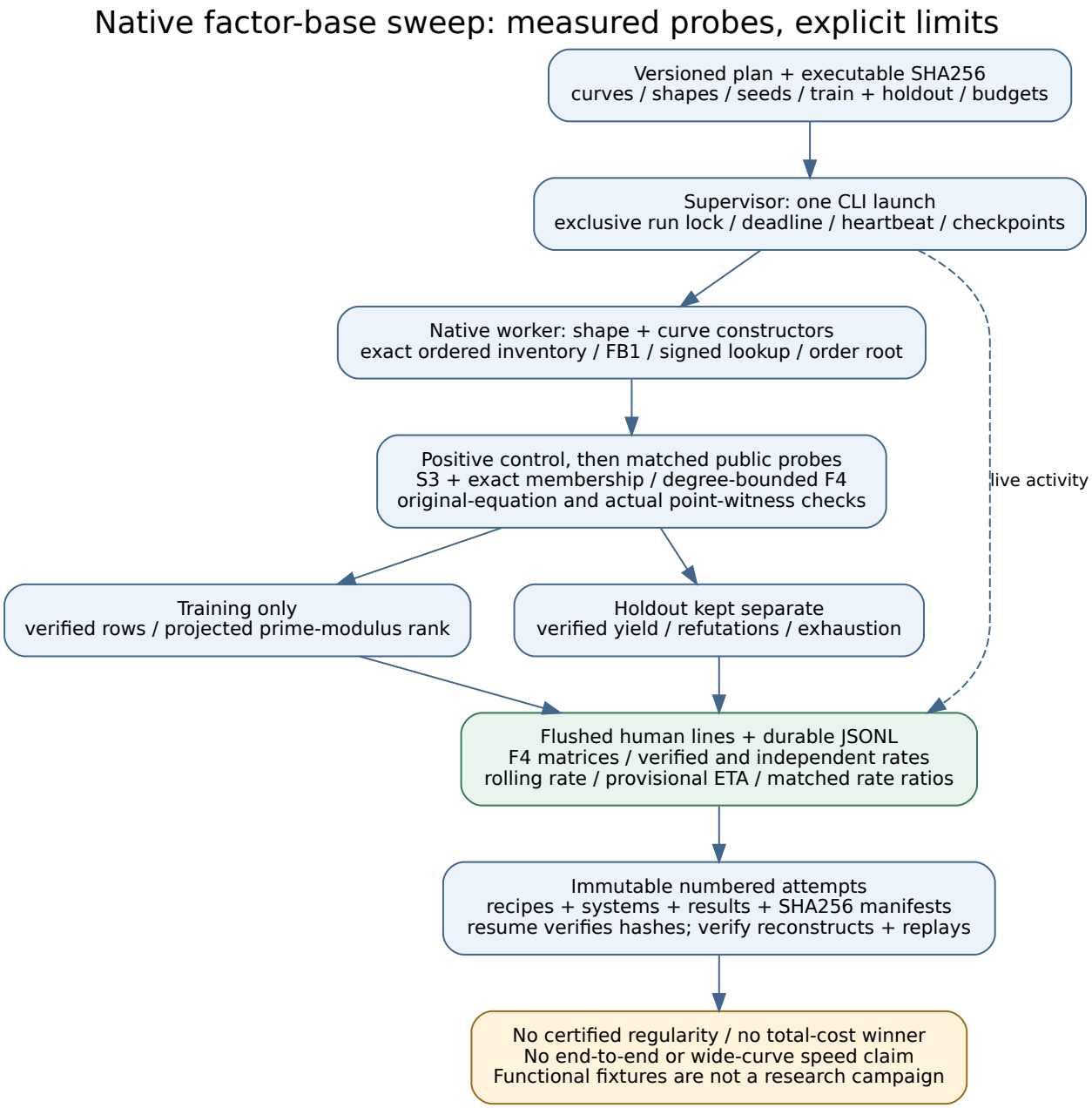
The solver path uses two summands. Prime systems use direct S3 and finite-set membership with coefficient modulus at most one million. Binary shape construction is bounded to field degree 31 and dimension 8. Wide 83/130-bit curves, general nonlinear binary inventories, other summand counts, algebraic-closure certificates and certified degree of regularity are not implemented. Exhaustion does not establish nonexistence. Successful solving degree is an observed procedural diagnostic, not certified regularity.

Actual validation

Native library	4,380 passed; four unchanged upstream failures; 117 existing ignores, in one complete serial command. No full-pass claim.
CLI unit / integration	Five unit and six executable integration tests passed. Retained run: 15 completed, one inapplicable; 16 sealed attempts and 29 actual point witnesses verified.
Affected native checks	All-targets compile, strict Clippy for the three touched bins, and four existing progress tests passed. Canonical maintained lint has two unchanged-target blockers.
Historical trace replay	Two unit passes, two registration-guard failures. Complete 384-record replay blocked: 48 historical model rows are absent from current main's registry.
CI-scope helper	9 tests passed after minimal base integration repair.
Unavailable	Conductor absent; pinned workflow HTTP 404. Neither is a pass.

Source and artifact binding: Native CLI b3ed888964100f1bce4780a4c0500a399c7aee50; retained executable SHA256 dd9c1e582effe751777081fc4d7fc6ba1acd90e1696797ef210d20da58c3c3f0. Library inputs are separately bound to its exact retained binary..

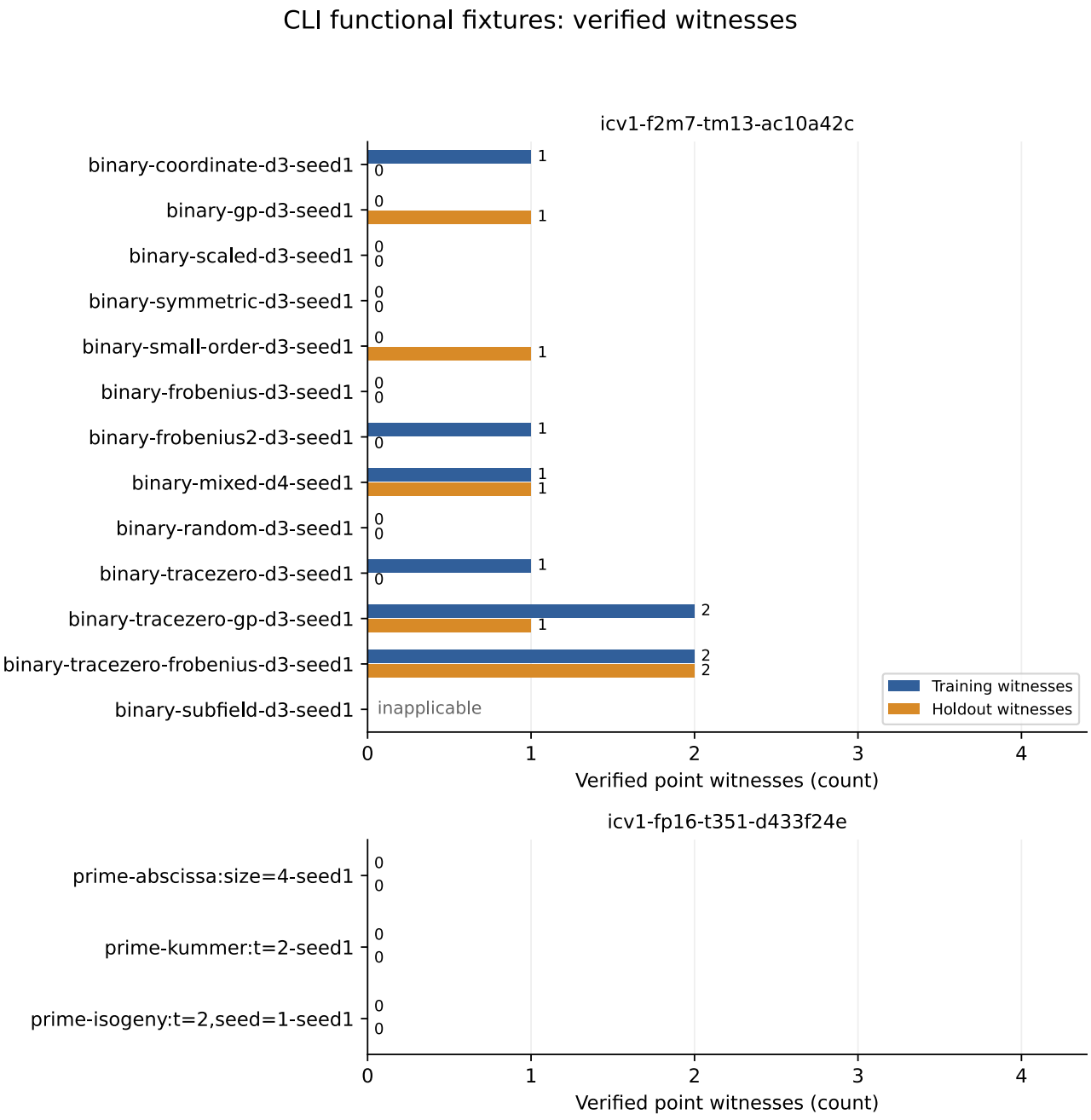
Execution and evidence boundaries



This diagram describes an implementation and functional checks, not a cryptanalytic performance result. No historical CPython generation or S3 maintenance operation ran. Frozen historical evidence, registries, aliases, stores and the draft, unrun, zero-funded planning card remain unchanged.

Sources: launch, plan, resume, verify and limitations requirements validation receipt. Editable visual source: architecture.dot.

Retained functional fixtures



Run functional-20261018T0025Z; per completed candidate: 4 training and 2 holdout probes.
Counts copied from native records; no uncertainty interval estimated. Missing/inapplicable results are not zero-yield observations.
Base sizes and parameterizations differ. This is a functional check, not a speed comparison or certified regularity result.

The native run completed 15 candidate configurations; one requested binary subfield was inapplicable. The completed cases retained 105 trials. Native verification checked all 16 sealed attempts and 29 actual point witnesses. Positive controls are separate from the plotted training/ holdout counts. No regularity, best-shape or speed conclusion follows from these small samples.

Source: recorded native counts rendering source.

Integration blockers and preservation

The full library command failed three unchanged trait-query cases because the committed traits JSON lacks the required `levels` field, plus one unchanged Jacobian classification expectation (`Boundary` versus `FormulaMarginal` for c2pnb304w1). The exact traits literal is present in the retained unit binary, with all 731 library inputs matched to source.

Historical native trace replay is blocked by the registration guard: all 48 model rows in the frozen historical source are missing from main’s current 2,375-row registry, while their alias targets remain. Two of four old trace unit tests therefore fail; no complete 384-record replay is claimed here.

The naming check reports 4,042 problems on unchanged inputs. Relevant ancestor pin files were checked; none was missing. Canonical maintained Clippy flags the unchanged research driver at line 206 (`drop_non_drop`) and the unchanged `n83_precomputed_rho` example at line 249 (`manual_is_multiple_of`). No assertion, resource limit or coverage condition was relaxed. Conductor remains absent, and its pinned workflow was HTTP 404; those checks are unavailable, not passed.

An environment restart interrupted earlier work. Partial and failed logs are retained alongside complete retries; unknown exits are not promoted to passes. Before the feature could build, a separate prerequisite commit repaired merged native declaration/import/dispatch defects, preserving both quartic families. All new library regression cases pass.

This feature leaves main’s frozen research, backlog, registry, aliases and store unchanged. That preserves scope; it does not claim to restore already-missing upstream registry rows. Historical CPython generation was not rerun. No new point logarithms, native performance claim, live S3 verification, paid host or research campaign is reported. The planning card remains draft, unrun, zero-funded and unleased.

Canonical scoreboard, leaderboard, timeline, performance-gain and figure paths were reviewed and left unchanged: there is no accepted new performance or mathematical result to place on them. Future measured comparisons retain the symmetric native-pricing bands and matched fastest external-reference gate.

Full details and source hashes: VALIDATION.md. Raw commands, failures, retries and input hashes: docs/ic/factor-base-sweep/validation/runs.tar.xz; commands, source manifests, complete/partial/failed logs, native attempt artifacts and recovery notes.. All 567 raw member hashes pass after extraction; deterministic repacking is byte-identical. Native verification of the extracted run also checks all 16 sealed attempts and 29 actual point witnesses. These integrity checks leave the recorded failures unchanged. See bundle receipts.